Topic 8:Energy Resource



Global Energy Statistics 2016

Total energy Production	China (2640 Mtoe)
Total energy Consumption	China (3101 Mtoe)
Crude oil Production	Saudi Arabia (565 Mt)
Oil products Production	USA (868 Mt)
Oil products Consumption	USA (751Mt)
Natural Gas Production	USA (769 bcm)
Natural Gas Consumption	USA (783 bcm)
Coal, lignite Production	China (3538 Mt)
Coal, lignite Consumption	China (3732 Mt)
Electricity Production	China (5682 TWh)
Electricity Consumption	China (4921 TWh)



Types of energy Resources

- NON- RENEWABLE SOURCE
- RENEWABLE SOURCE

NONRENEWABLE RESOURCES





A nonrenewable resource is a natural resource that cannot be re-made or re-grown at a scale comparable to its consumption.

FOSSIL FUEL



What are Fossil fuels?

- Fossil fuels are nonrenewable hydrocarbon-containing materials formed by the anaerobic decomposition of buried dead organisms.
- Fossil fuels are of great importance because they can be burned and produce significant amounts of energy.

Types of fossil fuel

Coal, oil (petroleum), and natural gas are considered nonrenewable because they can not be replenished in a short period of time. These are called fossil fuels.





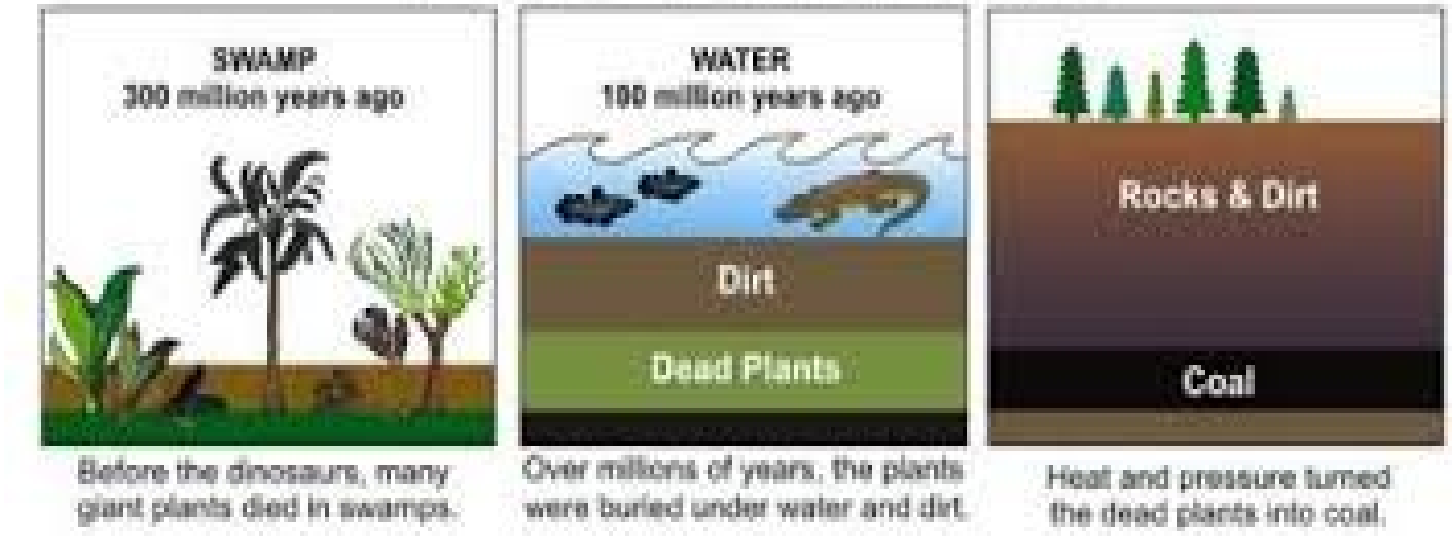


Coal Facts

- Most abundant fossil fuel produced in the United States.
- A nonrenewable energy source because it takes millions of years to create.
- Energy in coal comes from the energy stored by plants that lived hundreds of millions of years ago.

Formation of coal

- Formed when peat is altered physically and chemically.
- Formed from the remains of vegetation that grew as long as 400 million years ago.
- Process is called "coalification."



Oil facts



- Formed more than 300 million years ago and it has been used for more than 5,000-6,000 years.
- Scientists say that tiny diatoms (diatoms- sea creatures which can convert sunlight directly into stored energy) are the source of oil.
- Is produced in: Russia, Saudi Arabia, United States.

Formation of oil

- Oil was formed from the remains of animals and plants(diatoms) that lived millions of years ago in a marine (water) environment before the dinosaurs.

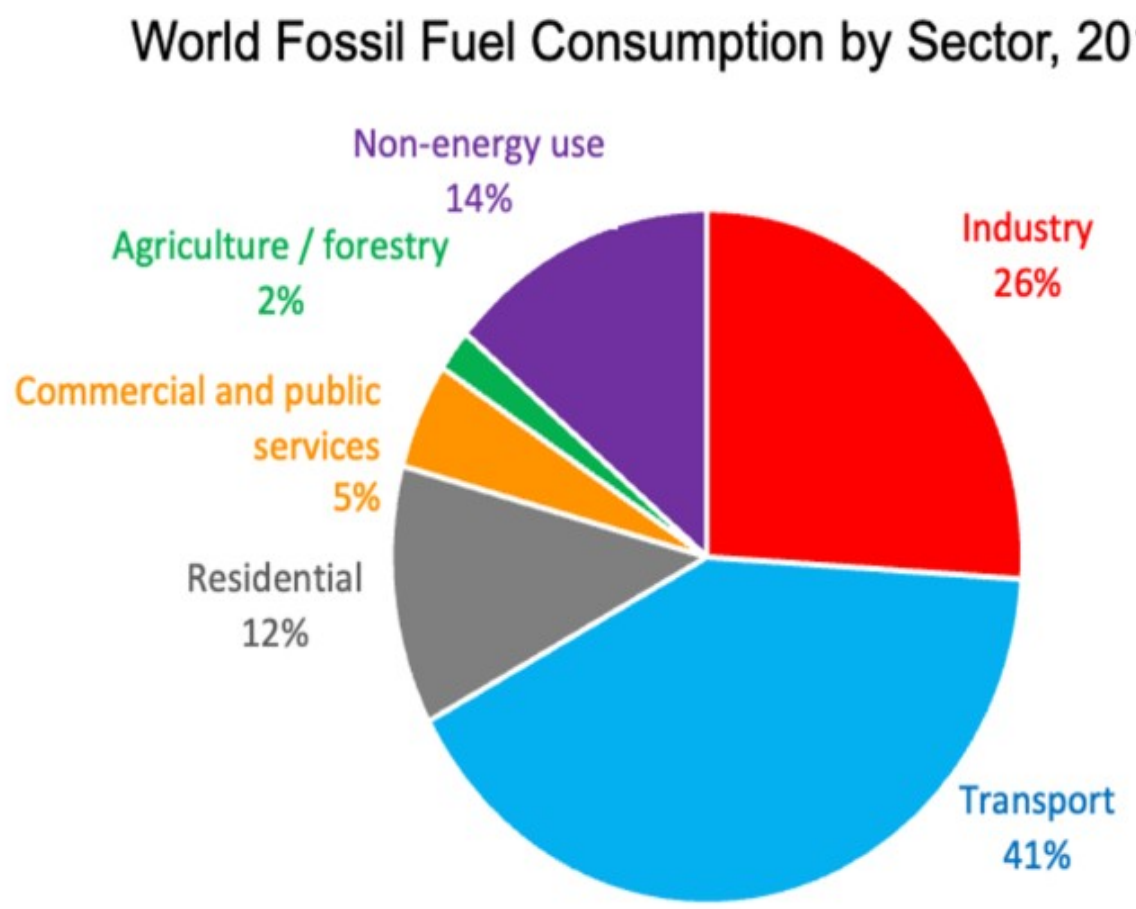
Natural gas facts

- Lighter than air , no odor.
- First discoveries of natural gas seeps were made in Iran.
- Mostly made up of a gas called methane.
- Usually found near petroleum underground and pumped from below ground and travels in pipelines to storage areas.

Formation of natural gas

- Methane, a gas or compound that has one carbon atom and four hydrogen atoms.

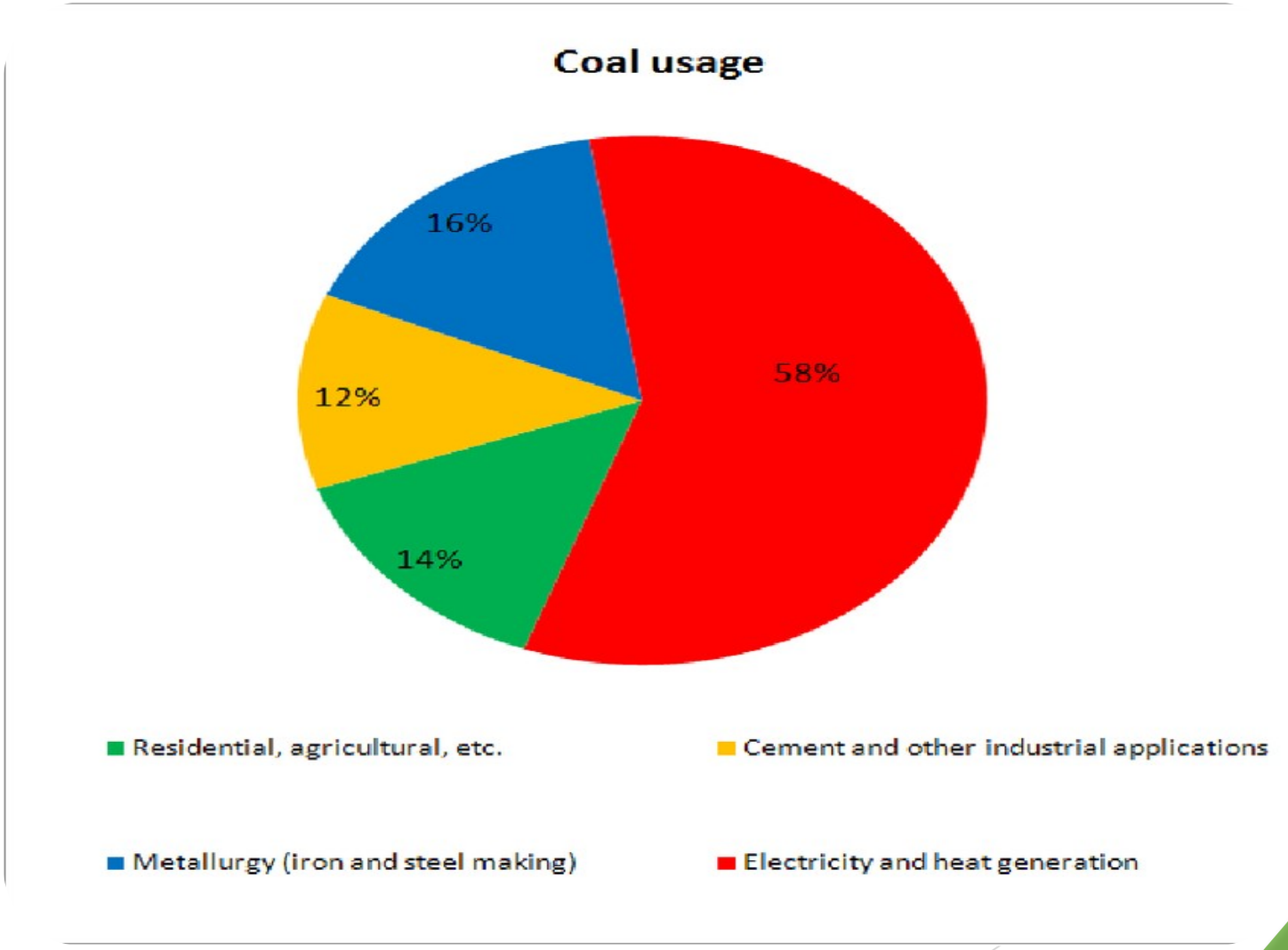
Fossil Fuel Consumption sector



Uses of Coal

- Industries.
- Domestic Use.
- Gasification and Liquefaction.
- Almost 93% of the coal used in the United States is used for generating electricity.

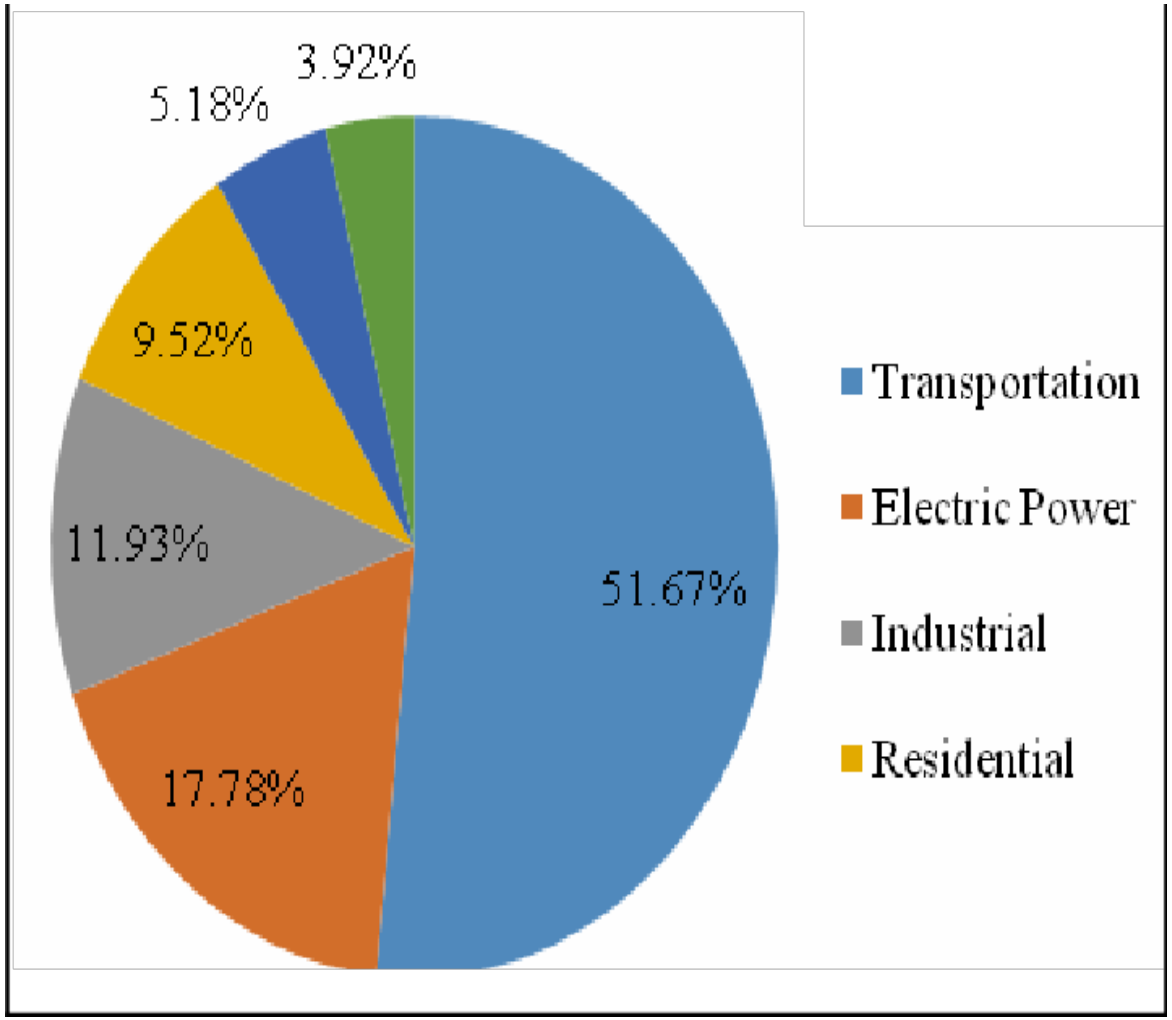
USES OF COAL


Coal usage
16%
12%
58%
14%
Residential, agricultural, etc.
Cement and other industrial applications
Metallurgy (iron and steel making)
Electricity and heat generation

Uses of Oil

- To propel vehicles
- To produce electricity
- In industrial areas
- For residential use

USES OF OIL


3.92%
5.18%
9.52%
11.93%
51.67%
17.78%
Transportation
Electric Power
Industrial
Residential

Uses Of Natural Gas

- Raw material to produce chemicals, fertilizer, and hydrogen.
- To heat the building and to dry clothes.
- To operate refrigeration and cooling equipment.
- For transportation services.



Disadvantages of fossil fuels

- Contribute to climate change.
- Non-renewable.
- Unsustainable.
- Hard to extract.
- Accident-prone.

Exxon valdez oil

- Unprecedented underwater leak spreads oil across the northern Gulf of Mexico between the mouth of the Mississippi River and Florida.
- Coast Guard says remote underwater cameras detect the well is leaking 1,000 barrels of crude oil per day.
- The Coast Guard says the leak will affect the Gulf shore.
- U.S. officials closed areas affected by the spill to fishing for an initial period of 10 days.

Deepwater Horizon

- The Deepwater Horizon oil rig exploded in the Gulf of Mexico
- There is some progress cleaning it up
- The last update showed that it was getting closer to the coast of Louisiana.
- Reach more than half the size of the Valdez disaster.
- More than 1 million gallons of water have been cleaned of oil, and 367881 Feet of oil booms have been deployed.
- BP has been trying all sorts of engineering efforts to stop the leak, including the four-story high containment dome, which was to be placed over the leak.|

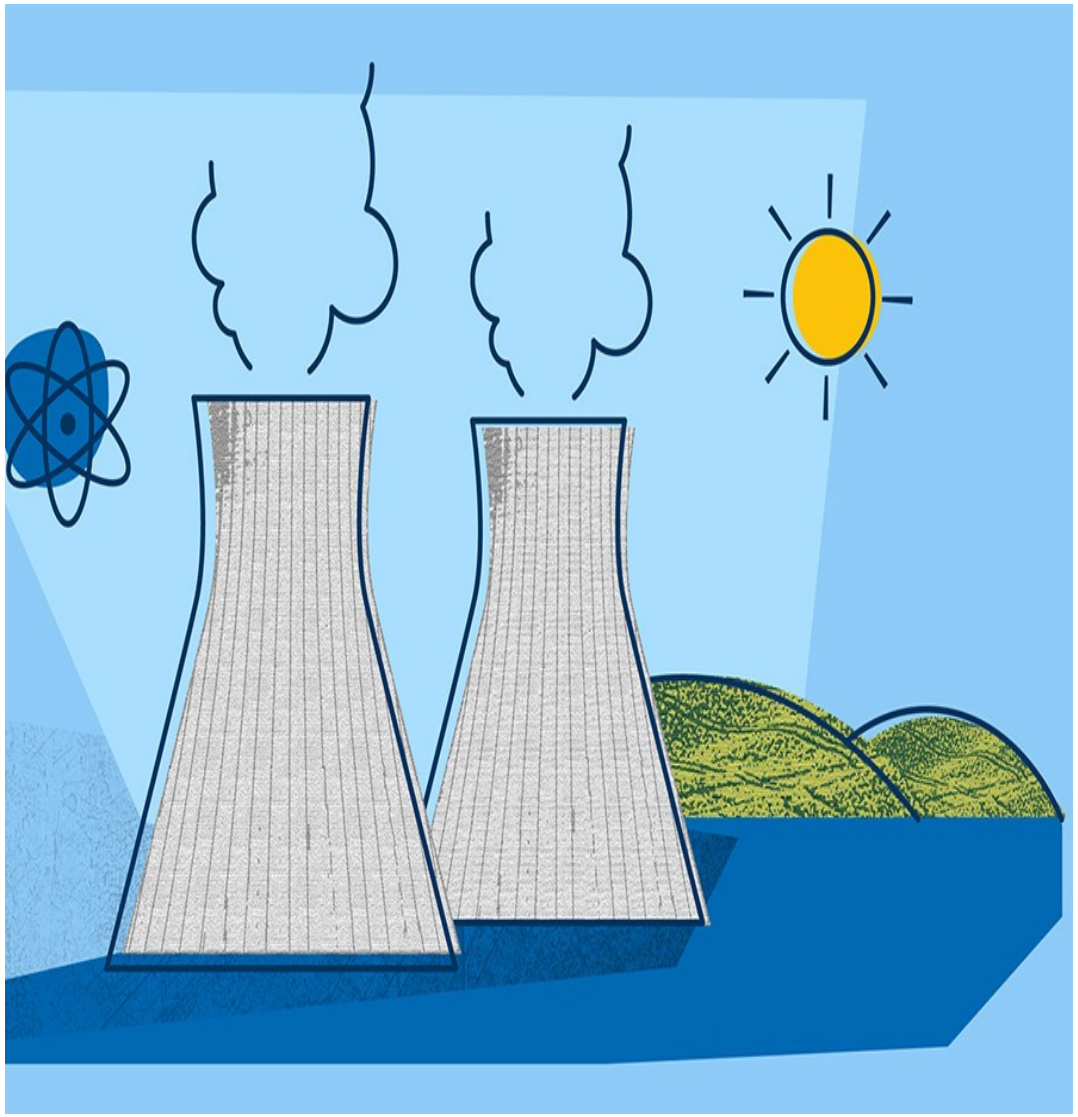
NUCLEAR ENERGY





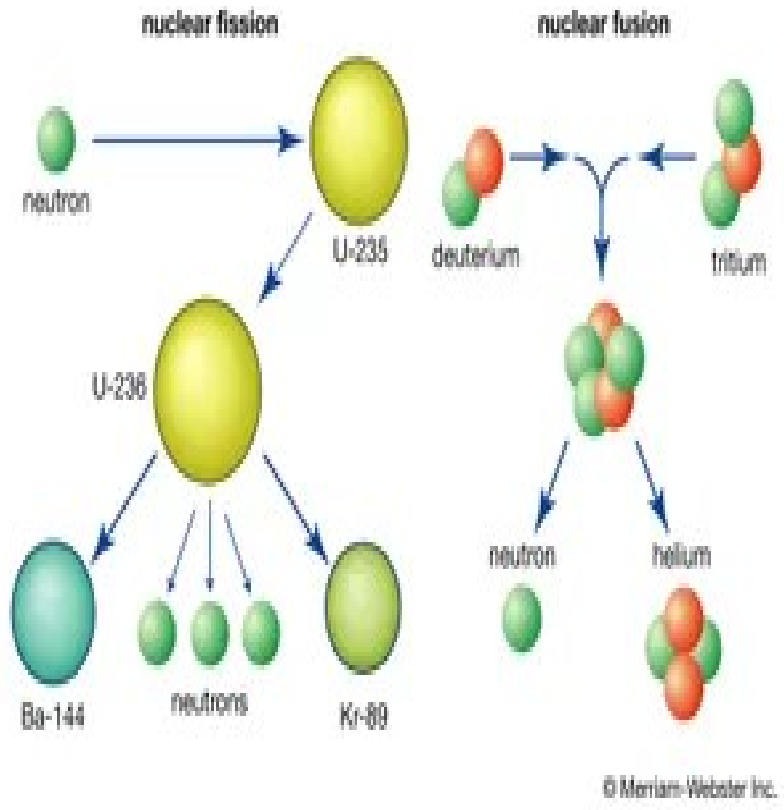
Nuclear fission and fusion use uranium to create energy.

Nuclear energy is a nonrenewable resource because once the uranium is used, it is gone!

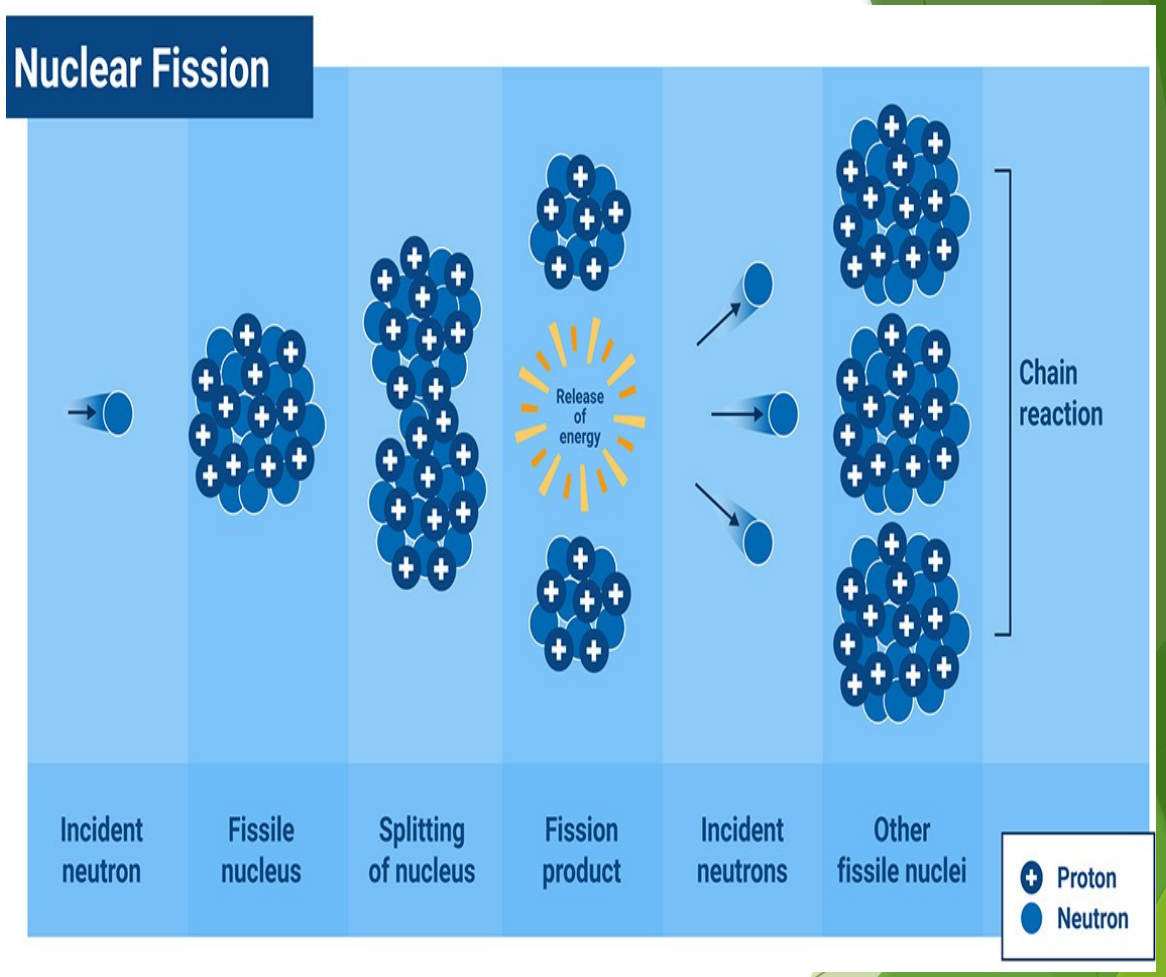


- Nuclear energy is a form of energy released from the nucleus, the core of atoms, made up of protons and neutrons. This source of energy can be produced in two ways: fission, when nuclei of atoms split into several parts, or fusion, when nuclei fuse(join or blend to form a single entity).
- The nuclear energy harnessed around the world today to produce electricity is through nuclear fission, while technology to generate electricity from fusion is at the R&D phase.

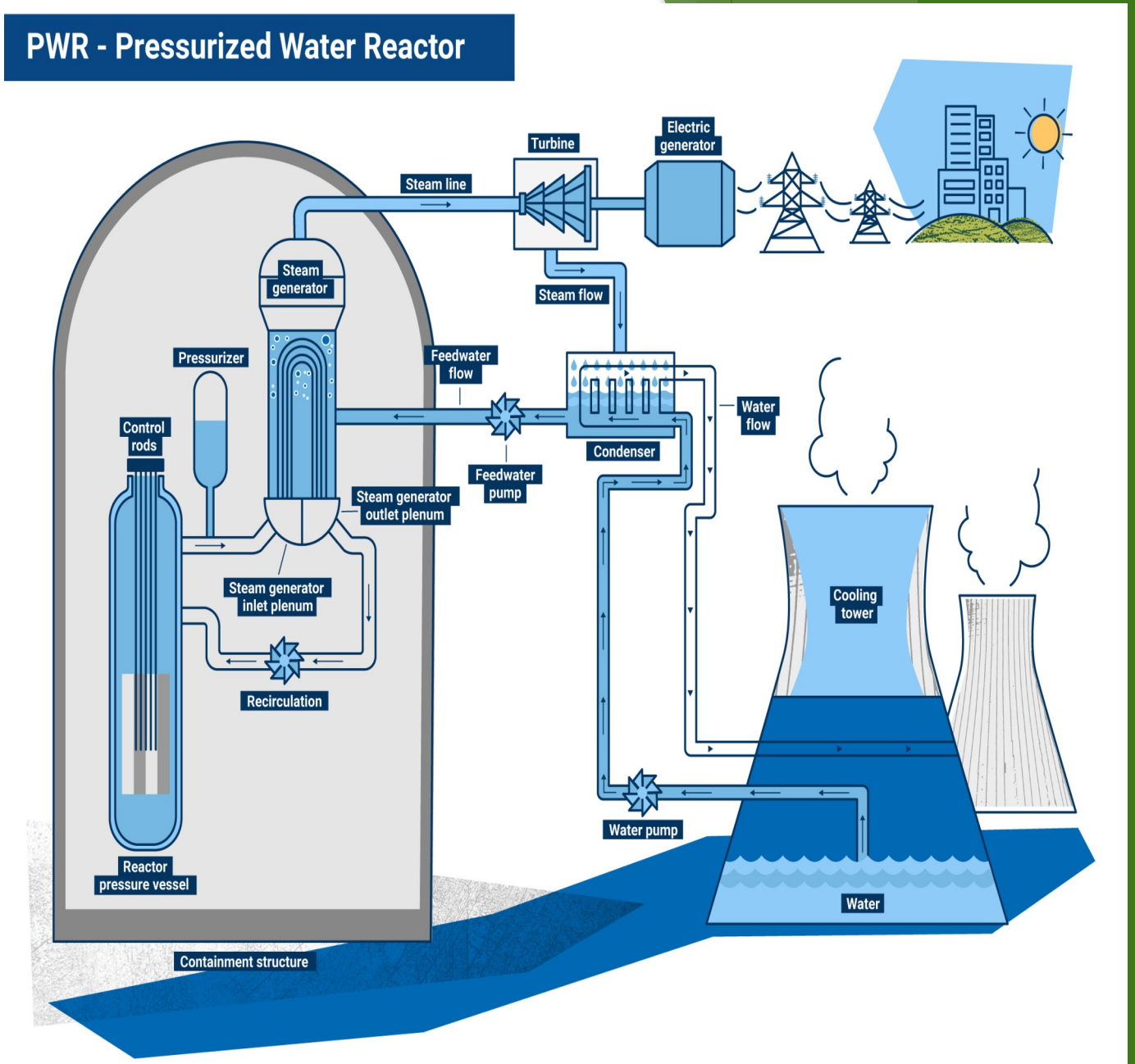
FISSION AND FUSION OF NUCLEAR ATOM



- In nuclear fission, for instance, when hit by a neutron, the nucleus of an atom of uranium-235 splits into two smaller nuclei, for example, a barium nucleus and a krypton nucleus, and two or three neutrons.
- These extra neutrons will hit other surrounding uranium-235 atoms, which will also split and generate additional neutrons in a multiplying effect, thus generating a chain reaction in a fraction of a second. Each time the reaction occurs, there is a release of energy in the form of heat and radiation. The heat can be converted into electricity in a nuclear power plant, similarly to how heat from fossil fuels such as coal, gas, and oil is used to generate electricity.
- Nuclear fusion is the process by which nuclear reactions between light elements form heavier elements. In cases where the interacting nuclei belong to elements with low atomic numbers (e.g., hydrogen [atomic number 1] or its isotopes deuterium and tritium), substantial amounts of energy are released.
- The vast energy potential of nuclear fusion was first exploited in therm onuclear weapons, or hydrogen bombs, which were developed in the decade immediately following World War II.



- The process may take place spontaneously in some cases or may be induced by the excitation of the nucleus with a variety of particles (e.g., neutrons, protons, deuterons, or alpha particles) or with electromagnetic radiation in the form of gamma rays.
- In the fission process, a large quantity of energy is released, radioactive products are formed, and several neutrons are emitted. These neutrons can induce fission in a nearby nucleus of fissionable material and release more neutrons that can repeat the sequence, causing a chain reaction in which many nuclei undergo fission, and an enormous amount of energy is released.
- If controlled in a nuclear reactor, such a chain reaction can provide power for society's benefit. If uncontrolled, as in the case of the so-called atomic bomb, it can lead to an explosion of awesome destructive force.



An example of a nuclear Reactor

- Inside nuclear power plants, nuclear reactors and their equipment contain and control the chain reactions, most fueled by uranium-235, to produce heat through fission. The heat warms the reactor's cooling agent, typically water, to produce steam. The steam is then channeled to spin turbines, activating an electric generator to create low-carbon electricity.

Advantages of Nuclear Energy

- Clean Energy Source: in the United States. It generates nearly 775 billion kilowatt-hours of electricity each year and produces nearly half of the nation's emissions-free electricity. This avoids more than 471 million metric tons of carbon each year, which is the equivalent of removing 100 million cars off the road.

- Creates Jobs: The nuclear industry supports nearly half a million jobs in the United States. Domestic nuclear power plants can employ up to 800 workers with salaries that are 50% higher than those of other generation sources. They also contribute billions of dollars annually to local economies through federal and state tax revenues.

- Supports National Security: A strong civilian nuclear sector is essential to U.S. national security and energy diplomacy. The United States must maintain its global leadership in this arena to influence the peaceful use of nuclear technologies. The U.S. government works with countries in this capacity to build relationships and develop new opportunities for the nation's nuclear technologies.

Challenges of Nuclear Energy

- Public Awareness: Commercial nuclear power is sometimes viewed by the public as a dangerous or unstable process. This perception is often based on three global nuclear accidents, its false association with nuclear weapons, and how it is portrayed on popular television shows and films. DOE and its national labs are working with industry to develop new reactors and fuels that will increase the overall performance of these technologies and reduce the amount of nuclear waste that is produced.
- Used Fuel Transportation, Storage, and Disposal: Many people view used fuel as a growing problem and are apprehensive about its transportation, storage, and disposal. DOE is responsible for the eventual disposal and associated transport of all used fuel, most of which is currently securely stored at more than 70 sites in 35 states. For the foreseeable future, this fuel can safely remain at these facilities until a permanent disposal solution is determined by Congress. DOE is currently evaluating nuclear power plant sites and nearby transportation infrastructure to support the eventual transport of used fuel away from these sites. Subject to appropriations, the Department is moving forward on a government-owned consolidated interim storage facility project that includes rail transportation.
- Constructing New Power Plants: Building a nuclear power plant can be discouraging for stakeholders. Conventional reactor designs are considered multi-billion-dollar infrastructure projects. High capital costs, licensing and regulation approvals, coupled with long lead times and construction delays, have also deterred public interest. microreactors image DOE is rebuilding its nuclear workforce by supporting the construction of two new reactors at Plant Vogtle in Waynesboro, Georgia. The units are the first new reactors to begin construction in the United States in more than 30 years. The expansion project supported up to 9,000 workers at peak construction and created 800 permanent jobs at the facility when the units came online in 2023 and 2024. DOE is also supporting the development of smaller reactor designs, such as microreactors and small modular reactors, that will offer even more flexibility in size and power capacity to the customer.
- High Operating Costs: Challenging market conditions have left the nuclear industry struggling to compete. DOE's Light Water Reactor Sustainability (LWRS) program is working to overcome these economic challenges by modernizing plant systems to reduce operation and maintenance costs while improving performance. In addition to its materials research that supports the long-term operation of the nation's fleet of reactors, the program is also looking to diversify plant products through non-electric applications such as water desalination and hydrogen production. To further improve operating costs. DOE is also working with industry to develop new fuels and cladding known as accident-tolerant fuels. These new fuels could increase plant performance, allowing for longer response times and producing less waste.

ALTERNATIVE OF FOSSIL FUEL



- Renewable Energy.
- Nuclear Power.
- Hydrogen.
- Biomass.
- Geothermal energy.

References

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